

A BEGINNER'S MANUAL

Autonomous AI Trading

Build, test, and run a hands-off trading bot, step by step

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GUIDES FOR EVERYTHING

Autonomous AI Trading

Build, test, and run a hands-off trading bot, step by step

This manual walks you, one small step at a time, from "I have never written code" to a **trading bot that runs by itself**: it wakes up on a schedule, reads market data, decides, checks its own safety rules, places orders through a broker's API, writes a log, and sends you a daily report. You'll build it in Python with an AI assistant (Claude Code) as your pair programmer, test it on years of history, and run it on **paper money** for weeks before a single real dollar is involved.

It is also an honest book. Most people who trade actively lose money, and no bot, AI model, or course can guarantee a profit. You'll see the real studies, learn the legal and tax basics, and build the guardrails (position limits, stop-losses, daily loss limits and a kill switch) that keep a mistake small. The goal is a calm, well-tested system you understand, sized so that even its worst day is one you can shrug off.

HOW TO READ THIS MANUAL

- **Tip** — a shortcut or good habit.
- **Note** — useful background.
- **Warning** — easy to get wrong.
- **Danger** — can lose data, money, or access.
- `$ command` — text you type, exactly as shown.
- Settings › Privacy — a menu path to click.

New words appear in **bold** the first time and are collected in the Glossary at the back.

This manual is education, not financial, legal, or tax advice. Trading involves risk, including the loss of all money you put in; past and simulated results do not predict future returns. Only trade money you can afford to lose, and paper-trade first. Broker features, fees, rules, and tax law were checked against official sources in October 2026 and change often; confirm current details with your broker, the IRS, FINRA, the SEC, or a qualified professional before acting.

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CHAPTER 1

Market Data

Your bot can only be as good as the prices it sees. **Market data** is the stream of prices and trading volume coming out of the exchanges. In this chapter you'll download your first prices, understand what they mean, and learn how data goes wrong.

IN THIS CHAPTER

- Bars, OHLCV, and timeframes
- Feeds: IEX versus SIP
- Downloading daily bars with `alpaca-py`
- Adjusted prices, missing data, and sanity checks

1.1 Bars and OHLCV

Prices are usually grouped into **bars** (also called candles). A daily bar summarises one trading day with five numbers, known as **OHLCV**:

FIELD	MEANING
Open	First trade price of the period
High	Highest price
Low	Lowest price
Close	Last trade price
Volume	How many shares traded

The bot in this manual uses **daily closes**: one number per day per symbol.

1.2 Feeds: IEX and SIP

U.S. stocks trade on many exchanges at once. The **SIP** (securities information processor) combines all of them. **IEX** is a single exchange. Alpaca's free Basic plan gives you real-time IEX data, while for historical requests it can serve full SIP data as long as you don't ask for the most recent 15 minutes. For a daily bot that reads *yesterday's* closes, the free plan is plenty.

1.3 Your first data script

Create a file named `hello_data.py`:

HELLO_DATA.PY

```
import os
from datetime import datetime, timedelta, timezone

from dotenv import load_dotenv
from alpaca.data.historical import StockHistoricalDataClient
from alpaca.data.requests import StockBarsRequest
from alpaca.data.timeframe import TimeFrame
from alpaca.data.enums import Adjustment

load_dotenv()
client = StockHistoricalDataClient(
    os.environ["ALPACA_API_KEY"], os.environ["ALPACA_SECRET_KEY"]
)

request = StockBarsRequest(
    symbol_or_symbols=["SPY", "TLT"],
    timeframe=TimeFrame.Day,
    start=datetime.now(timezone.utc) - timedelta(days=30),
    adjustment=Adjustment.ALL,
)
bars = client.get_stock_bars(request).df
print(bars.tail(10))
```

Run it with `python hello_data.py`. You'll see a table with a two-level index (symbol, then timestamp) and columns including `open`, `high`, `low`, `close`, and `volume`.

What each part does:

- `StockHistoricalDataClient` is the object that talks to Alpaca's data service.
- `StockBarsRequest` describes what you want: which symbols, what bar size, and from when.
- `Adjustment.ALL` asks for prices **adjusted** for stock splits and dividends (explained below).
- `.df` turns the answer into a pandas **DataFrame**, a table you can calculate with.

Reshape into one column per symbol

Strategies are easier to write when each symbol is its own column:

PYTHON

```python
closes = bars["close"].unstack(level=0)   # rows: dates, cols: symbols
closes.index = closes.index.date          # drop the time of day
print(closes.tail())
```

Crypto data

Alpaca's crypto data doesn't even need keys:

PYTHON

```python
from alpaca.data.historical import CryptoHistoricalDataClient
from alpaca.data.requests import CryptoBarsRequest

crypto = CryptoHistoricalDataClient()
req = CryptoBarsRequest(
    symbol_or_symbols=["BTC/USD"],
    timeframe=TimeFrame.Day,
    start=datetime.now(timezone.utc) - timedelta(days=30),
)
print(crypto.get_crypto_bars(req).df.tail())
```

Note the slash: Alpaca writes crypto pairs as `BTC/USD`.

1.4 Adjusted prices

When a company does a 4-for-1 **stock split**, the price drops to a quarter overnight, but nobody lost money. Unadjusted data shows a fake 75% crash, and your bot might sell in panic. **Adjusted** data rewrites history to remove that jump, and also accounts for **dividends** (cash payments to shareholders). Always backtest and signal on adjusted prices.

1.5 How data goes wrong

PROBLEM	SYMPTOM	DEFENSE
Missing days	Fewer rows than expected	Require a minimum row count
Today's partial bar	Last "close" is a mid-day price	Drop today's row before computing signals
Bad tick	One absurd price	Reject moves over, say, 25% in a day for an ETF
Stale data	Last date is days old	Check the last date is the previous trading day
Rate limits	HTTP 429 errors	Fewer requests; ask for many symbols at once

A quick sanity check you'll reuse in the bot:

PYTHON

```python
def sane(series, max_jump=0.25, min_rows=200):
    """True if a price series looks usable."""
    s = series.dropna()
    if len(s) < min_rows:
        return False
    biggest_move = s.pct_change().abs().max()
    return biggest_move < max_jump
```

> ⚠ **WARNING**
>
> Free, unofficial data sources such as the `yfinance` library are handy for quick experiments, but they scrape public websites, can break without warning, and aren't licensed for every use. For anything your bot trades on, use your broker's official data.

1.6 Other data you might want

- **News:** Alpaca's data API includes news articles for stocks and crypto (used in Chapter 10).
- **Market calendar and clock:** Alpaca's trading API tells you if the market is open and lists trading days and early closes (Chapter 16).
- **Fundamentals and alternative data:** Company earnings, sentiment, and so on. Skip these for your first bot.

- [ ] I ran `hello_data.py` and saw real daily bars
- [ ] I can reshape bars into one column per symbol
- [ ] I use adjusted prices for anything my bot decides with
- [ ] I know the five ways data goes wrong and how to check for them

Glossary

Adjusted prices	Historical prices rewritten to remove the effect of stock splits and dividends.
AI agent	An AI that can take actions, such as placing orders, not just answer questions.
API	Application programming interface: a doorway that lets one program talk to another, such as your bot talking to your broker.
API key	A pair of secret codes (key ID and secret) that lets a program act on your account.
Atomic write	Saving a file by writing a temporary copy and swapping it in, so the file is never half-written.
Backtest	Replaying a strategy on historical prices to see how it would have performed.
Bar	A summary of trading over a period (open, high, low, close, volume). Also called a candle.
Benchmark	What you compare a strategy with, usually buying and holding the same fund.
Bracket order	An entry order plus a take-profit and a stop-loss that activate when it fills.
Broker	A company that holds your investment account and sends your orders to the market.
CAGR	Compound annual growth rate: the average yearly return, with compounding.
Cash account	A brokerage account where you can only buy with deposited money; no borrowing.
Circuit breaker	A rule that halts the bot when results suggest something is wrong, such as a large drawdown.

Client order ID	Your own label attached to an order, so you can trace it back to a specific run.
cron	The classic Linux scheduler that runs commands at set times.
Crypto exchange	A company that runs a marketplace for cryptocurrencies, usually holding your coins.
CSV	Comma-separated values: a plain-text table that opens in any spreadsheet.
Daily loss limit	A rule that stops trading for the day after the account falls a set percentage.
Dataclass	A Python container for a group of related settings or values.
DataFrame	A pandas table of rows and columns, used for prices and calculations.
Day trading	Opening and closing positions within the same trading day.
Digital assets question	The yes/no question about crypto on U.S. federal tax returns.
Digital commodity	A crypto asset, such as bitcoin, that the SEC and CFTC treat as a commodity rather than a security.
Dividend	A cash payment a company or fund makes to its shareholders.
Dollar-cost averaging (DCA)	Buying a fixed dollar amount at regular intervals regardless of price.
Drawdown	The fall from an account's or strategy's highest value to a later low.
Dry run	Running the bot so it makes and logs decisions without sending any orders.
Equity curve	A chart or series showing how an account's value changed over time.
ETF	Exchange-traded fund: a basket of investments that trades like a single stock.
Form 1099-DA	The IRS form crypto brokers use to report your digital asset sales.
Fractional share	Part of one share, bought by specifying a dollar amount.
Free-riding	Buying and selling a security in a cash account before paying for it; can lead to a 90-day restriction.
Good faith violation	Selling a security bought with unsettled funds before those funds settle, in a cash account.
Grid bot	A bot that places buy and sell orders at evenly spaced prices to profit from price swings in a range.
GTC	Good-till-cancelled: an order that stays open until filled or cancelled (Alpaca cancels them after 90 days).

Guardrail	A rule that keeps a component's mistakes harmless, such as defaulting an unexpected AI answer to "neutral."
Heartbeat	A check that the bot actually ran, so silent failures are noticed.
IEX	Investors Exchange, a single U.S. stock exchange whose real-time data Alpaca provides free.
Intraday margin standards	FINRA's 2026 replacement for the pattern day trader rules, based on exposure during the day.
IP allowlist	A setting that only accepts an API key from listed internet addresses. Also called whitelisting.
Kill switch	A way to stop the bot and close all positions immediately.
Large language model (LLM)	An AI model that reads and writes text, such as Claude.
Least privilege	Giving each key or tool only the permissions it truly needs.
Leverage	Controlling a larger position than your deposit, magnifying gains and losses.
Liquidity	How easily something can be bought or sold without moving its price.
Look-ahead bias	A backtest error that uses information you wouldn't have had at the time.
Machine learning (ML)	Letting a program find patterns in data instead of writing rules by hand.
Maker / taker	A maker order waits on the order book and adds liquidity; a taker order fills immediately. Fees often differ.
Margin account	A brokerage account that lets you borrow to buy or sell short.
Market data	Prices and volumes coming from exchanges.
Market order	An order to buy or sell immediately at the best available price.
MCP	Model Context Protocol: a standard way to connect AI apps to outside tools such as a broker.
Mean reversion	A strategy betting that unusually stretched prices return toward their average.
Momentum	A strategy that buys recent relative winners, expecting them to keep outperforming for a while.
Moving average	The average price over the last N days, recalculated daily.
Notional order	An order specified in dollars rather than shares.

OHLCV	Open, high, low, close, volume: the five numbers in a price bar.
Order	An instruction to your broker to buy or sell.
OTO order	One-triggers-other: an entry order that activates a second order (such as a stop-loss) when it fills.
Overfitting	Tuning rules so closely to past data that they capture noise and fail on new data.
Paper trading	Simulated trading with fake money against real prices.
Pattern day trader (PDT)	A former FINRA designation for frequent day traders in margin accounts, with a $25,000 minimum; replaced in 2026.
Position	The amount of a particular investment you currently hold.
Position sizing	Deciding how much to buy, for example with the 1–2% risk rule.
Prompt injection	Hidden text that tries to trick an AI into doing something it shouldn't.
Pure function	A function that only uses its inputs and returns an answer, with no network or file access.
Rebalancing	Trading back to a target mix of investments when it drifts.
Risk rules	Code that can veto a trade, such as size limits and loss limits.
Rotating log	A log file that starts a new file at a size limit and keeps only a few old ones.
Sandbox	A test version of an API; some simulate trading, others return canned answers.
Settlement	When a trade officially completes and cash or shares change hands; T+1 for most U.S. stocks.
Sharpe ratio	Return divided by volatility; a measure of return per unit of risk.
Short selling	Selling borrowed shares to profit if the price falls; requires margin.
Short-term capital gain	Profit on an asset held one year or less; taxed as ordinary income in the U.S.
SIP	Securities information processor: the combined data feed of all U.S. exchanges.
SIPC	Securities Investor Protection Corporation: protects customers if a member brokerage fails, up to $500,000 including $250,000 cash.
Slippage	The difference between the price you expected and the price you got.
Spot	Buying the actual asset (such as a coin) rather than a contract about it.

Spread	The gap between the best buying and selling prices.
Standard deviation	A measure of how spread out values are around their average.
State file	A small file where the bot remembers things between runs.
Stop-loss	An order that sells automatically if the price falls to a chosen level.
Strategy	Written rules precise enough for a computer to follow.
Survivorship bias	Testing only on assets that still exist, ignoring those that failed.
systemd timer	A Linux scheduler built into Ubuntu that can catch up on missed runs.
T+1	Settlement one business day after the trade date.
Terminal	A window for typing commands instead of clicking.
Timeframe	How often the bot looks at prices and can trade (daily, hourly, and so on).
TOML	A simple settings file format of names and values.
Trend following	A strategy that holds assets while their prices are trending up.
Virtual environment	A private set of Python libraries for one project.
VM	Virtual machine: a rented computer in a data centre.
Walk-forward testing	Choosing parameters on past data, testing them on the next unseen period, and rolling forward.
Wash sale	Selling at a loss and buying substantially identical securities within 30 days before or after; the loss is deferred.
Watchlist	The short list of symbols the bot is allowed to trade.
Z-score	How many standard deviations a value is from its average.